

LOAN APPROVAL PREDICTION SYSTEM

Student Name: SAMEERA O S

Register Number: 40

Class: CSE-C

Project Title: Loan Approval Prediction System using Machine Learning

Subject: Python Individual Mini Project

ABSTRACT

The Loan Approval Prediction System is a machine learning-based application that predicts whether a loan application is likely to be approved or rejected. Financial institutions receive many loan requests, and evaluating them manually can be time-consuming. This project uses historical loan data and machine learning techniques to assist in decision making. The system takes inputs such as applicant income, loan amount, credit history, and employment status, then processes the data using a trained model to predict the loan approval status. The application is built using Python and implemented with Streamlit to create a simple web interface. Users can enter the required details through the interface and instantly receive a prediction result. This project demonstrates how machine learning can support banking systems by making faster and more reliable loan approval decisions.

PROBLEM DESCRIPTION

Loan approval is an important process in banking and financial institutions. Every day, banks receive a large number of loan applications from individuals and businesses. Traditionally, loan approval decisions are made by bank officials after carefully analyzing the applicant's financial information, credit history, and other supporting documents.

However, this manual evaluation process can be slow and sometimes inconsistent. Human errors or subjective judgments may also affect the decision-making process. With the growth of data and technology, machine learning can be used to assist banks in analyzing loan applications more efficiently.

The objective of this project is to develop a system that can predict whether a loan application will be approved or rejected based on applicant information. By analyzing historical loan data, the machine learning model learns patterns and relationships between different factors such as income, loan amount, and credit history. The system then uses this knowledge to make predictions for new loan applications.

This project demonstrates how machine learning can help financial institutions make faster and more consistent decisions.

LIBRARIES & TOOLS USED

Python

Python is the main programming language used to develop the project. It is widely used for data analysis, machine learning, and application development.

Pandas

Pandas is used for data manipulation and preprocessing. It helps in reading datasets, cleaning data, and preparing it for machine learning models.

NumPy

NumPy is used for numerical operations and handling arrays during data processing.

Scikit-learn

Scikit-learn is used to build and train the machine learning model. It provides algorithms for classification and prediction.

Joblib

Joblib is used to save and load the trained machine learning model. The trained model is stored in a file called **model.pkl** so that it can be reused without retraining.

Streamlit

Streamlit is used to create a simple and interactive web application interface where users can input loan details and view prediction results.

SYSTEM DESIGN / WORKFLOW

The workflow of the Loan Approval Prediction System consists of several steps.

1. **Data Collection**

Historical loan application data is collected and used for training the machine learning model.

2. **Data Preprocessing**

The dataset is cleaned by handling missing values and converting categorical variables into numerical format.

3. Model Training

A machine learning algorithm is trained using the processed dataset to learn patterns related to loan approval.

4. Model Saving

The trained model is saved using Joblib as **model.pkl** so that it can be used later for predictions.

5. User Interface

A web interface is created using Streamlit where users can input loan application details.

6. Prediction

When the user submits the details, the system processes the input and sends it to the trained model.

7. Result Display

The model predicts whether the loan will be approved or rejected, and the result is displayed on the screen.

The application runs locally and can be accessed through a browser at <http://localhost:8501>.

IMPLEMENTATION DETAILS

The implementation of the Loan Approval Prediction System is divided into two main stages: model development and application development.

First, the dataset containing previous loan applications is loaded using the Pandas library. The data is cleaned and preprocessed to ensure that it is suitable for training the machine learning model. Categorical values such as gender, education, and employment status are converted into numerical form so that the model can process them.

After preprocessing, the dataset is divided into training and testing sets. A machine learning algorithm from the Scikit-learn library is used to train the prediction model. The trained model learns relationships between applicant attributes and loan approval outcomes.

Once the model is trained, it is saved as a **model.pkl** file using Joblib. This allows the application to load the trained model quickly without retraining.

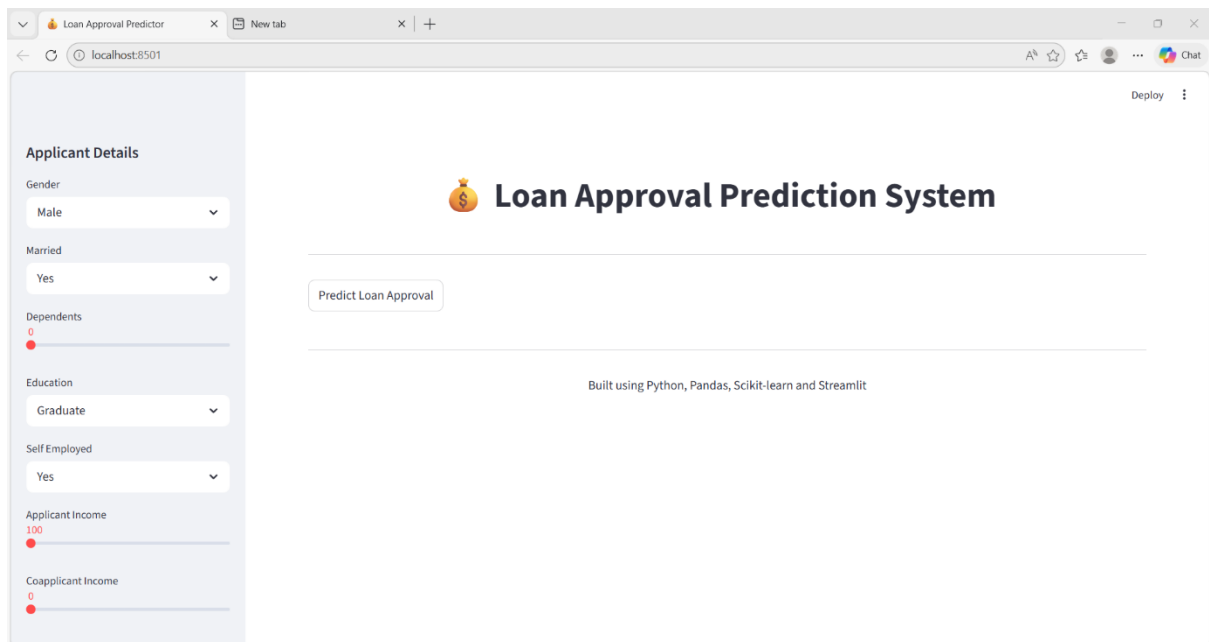
Next, a web application is developed using Streamlit. The interface allows users to enter information such as income, loan amount, credit history, and employment status. When the user submits the form, the input data is processed and passed to the trained

model. The model then predicts the loan approval status and displays the result on the screen.

RESULTS & OUTPUT

The application provides a simple interface where users can input loan details and receive predictions.

Screenshot 1 – Application Interface



The screenshot shows a web browser window with the title "Loan Approval Predictor". The address bar shows "localhost:8501". The interface is divided into two main sections. On the left is a sidebar titled "Applicant Details" containing several input fields: "Gender" (dropdown menu with "Male" selected), "Married" (dropdown menu with "Yes" selected), "Dependents" (slider bar with "0" selected), "Education" (dropdown menu with "Graduate" selected), "Self Employed" (dropdown menu with "Yes" selected), "Applicant Income" (slider bar with "100" selected), and "Coapplicant Income" (slider bar with "0" selected). On the right is the main content area, which features a large heading "Loan Approval Prediction System" with a money bag icon. Below the heading is a button labeled "Predict Loan Approval". At the bottom of the main content area, it says "Built using Python, Pandas, Scikit-learn and Streamlit". In the top right corner of the browser window, there is a "Deploy" button and a "Chat" icon.

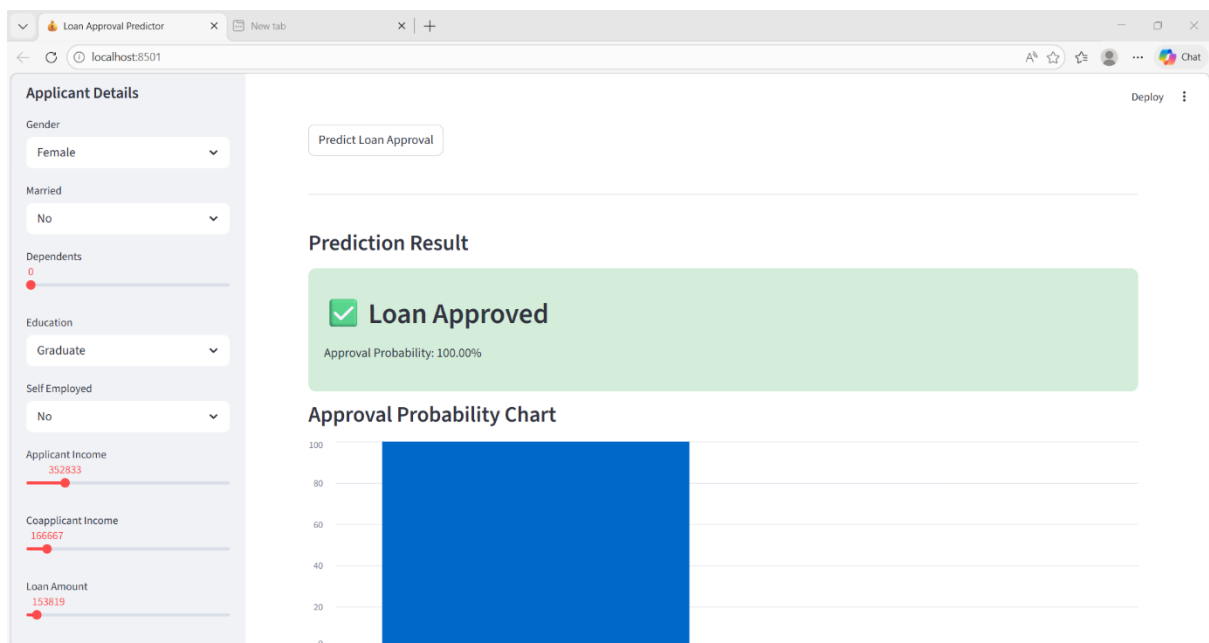
The Streamlit interface displays input fields where users can enter applicant details such as income, loan amount, and credit history.

Screenshot 2 – Processing Stage

The screenshot shows a web application titled "Loan Approval Prediction System" running on localhost:8501. The interface is divided into two main sections. On the left is a sidebar titled "Applicant Details" containing several input fields: Gender (Male), Married (Yes), Dependents (0), Education (Graduate), Self Employed (Yes), Applicant Income (360923), Coapplicant Income (0), and Loan Amount (315628). On the right is the main content area, which features a large heading "Loan Approval Prediction System" with a money bag icon, a "Predict Loan Approval" button, and a note stating "Built using Python, Pandas, Scikit-learn and Streamlit".

When the user clicks the prediction button, the application processes the input data and sends it to the machine learning model.

Screenshot 3 – Final Prediction Output



The system displays the result indicating whether the loan is likely to be **Approved** or **Rejected**.

CONCLUSION & SCOPE FOR MINI PROJECT EXTENSION

The Loan Approval Prediction System demonstrates how machine learning can be used to support decision-making in the banking sector. By analyzing applicant information and predicting loan approval outcomes, the system helps automate the initial screening process and reduce manual effort.

The project successfully integrates data preprocessing, machine learning model training, and web application development using Streamlit. The system provides quick predictions based on user input and demonstrates the practical application of machine learning in finance.

In the future, this project can be expanded by using larger datasets, improving model accuracy with advanced algorithms, and deploying the application online so that it can be accessed by real users.

REFERENCES

1. Scikit-learn Documentation – <https://scikit-learn.org>
2. Pandas Documentation – <https://pandas.pydata.org>
3. Streamlit Documentation – <https://streamlit.io>
4. Machine Learning resources and tutorials.